Antenna Arrays

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Lecture Objective

By the end of this lecture, you will be able to:

- Analyze the radiation pattern properties of an antenna array.

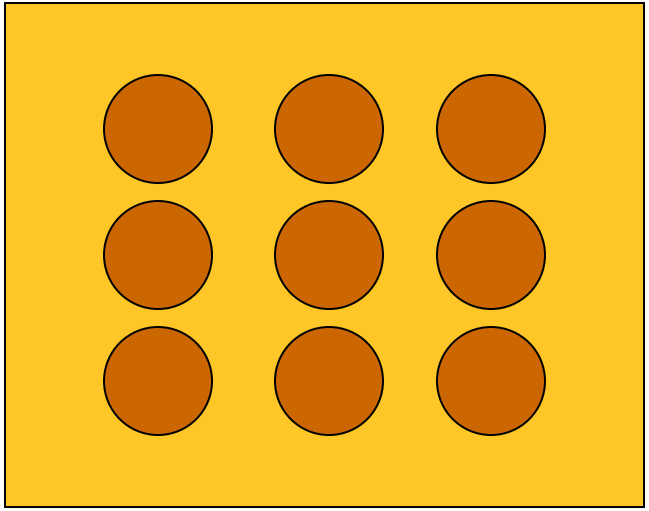
Overview of Antenna Arrays

- An effective way to achieve desired radiation characteristics (gain, sidelobe level, location of peaks and nulls) which cannot be obtained using a single antenna of any kind by itself is to use an antenna ***array***.

Definition of Antenna Array

- An antenna array is an electrical and geometrical arrangement of discrete radiating elements.
- The term “electrical” refers to the relative amplitudes and phases of the excitations of each element.
- The term “geometrical” refers to the relative positions of the elements.

Array Geometry



Top-view of a 3 X 3 array of circular patch elements

General Characteristics of Antenna Arrays (1 of 4)

- Usually (but not always), the individual radiating elements of an array are identical.
- Usually (but not always), the individual radiating elements of an array are on the order of $\lambda/2$ or smaller.
  - Two important examples of arrays comprising elements which are not identical are Yagi-Uda antenna and log-periodic dipole array (LPDA).

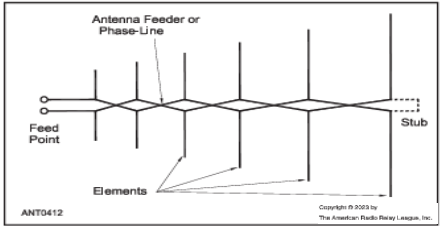


Figure 10.1 — The basic components of a log periodic dipole array (LPDA). The forward direction is to the left in this sketch. Many variations of the basic design are possible.

  - The Very Large Array (VLA) radio telescope comprises many electrically large reflector elements.

https://public.nrao.edu/vla-configurations/

General Characteristics of Antenna Arrays (2 of 4)

- The properties of an array are determined by
  - the shape of the overall array
  - the relative displacement between the elements
  - the excitation amplitude of the individual elements
  - the excitation phase of the individual elements
  - the relative patterns of the individual elements

General Characteristics of Antenna Arrays (3 of 4)

- The first four factors determine the array factor.
- The last factor is the element factor (pattern).

General Characteristics of Antenna Arrays (4 of 4)

- For large arrays of identical elements:

$$\boxed{\begin{aligned}\tilde{\vec{E}}(\text{total}) &= \left[\tilde{\vec{E}}(\text{single element at reference point})\right] \\ &\quad \times [\text{array factor}]\end{aligned}}$$

Electric Field Intensity of Individual Element

- For "reference" element:

Spherical wave factor

$$\tilde{E}_e(R,\theta,\phi) = \frac{e^{-jk_0R}}{R}\tilde{f}_e(\theta,\phi)$$

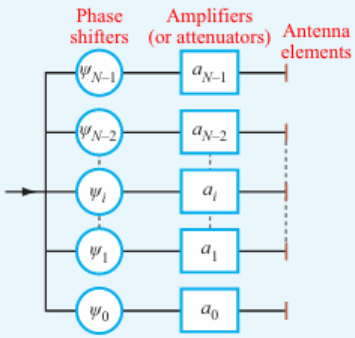
Element pattern

- Contribution to array field:

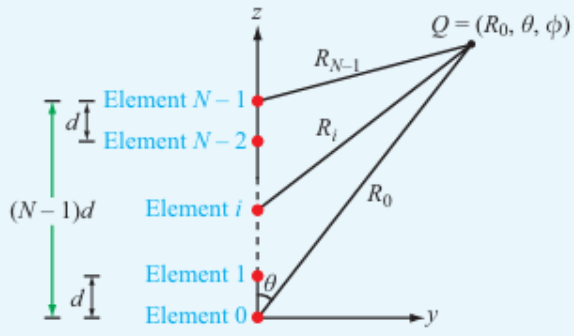
$$\tilde{E}_i(R,\theta,\phi) = a_i e^{j\psi_i}\frac{e^{-jk_0R_i}}{R_i}\tilde{f}_e(\theta,\phi)$$

Complex excitation coefficient

Linear Array Excitation and Geometry



(a) Array elements with individual amplitude and phase control



(b) Array geometry relative to observation point

Figure 9-26 Linear-array configuration and geometry.

Linear Array Geometry

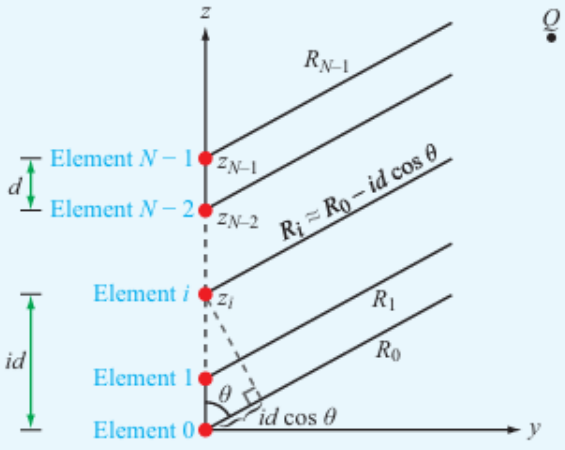


Figure 9-27 The rays between the elements and a far-away observation point are approximately parallel lines. Hence, the distance $R_i \approx R_0 - id\cos\theta$.

Assume Q is very far away compared to the length of the array, so we can use the ***parallel rays approximation:***

$$R_i \approx R_0 - z_i \cos\theta\,, \text{ for phase}$$
$$\approx R_0, \text{ for magnitude}$$

$$z_i = id$$

Electric Field Intensity and Radiation Power Intensity of Array

$$\tilde{E}(R_0,\theta,\phi) = \sum_{i=0}^{N-1} \tilde{E}_i(R,\theta,\phi) = \frac{e^{-jk_0R_0}}{R_0}\tilde{f}_e(\theta,\phi)\sum_{i=0}^{N-1} a_i e^{j\psi_i} e^{jikd\cos\theta}$$

$$U(\theta,\phi) = \frac{1}{2\eta_0}\left|\tilde{f}_e(\theta,\phi)\right|^2 \left|\sum_{i=0}^{N-1} a_i e^{j\psi_i} e^{jikd\cos\theta}\right|^2$$

Element Factor, Array Factor, and Pattern Multiplication

$$F(\theta,\phi) = \underbrace{\left|\tilde{f}_e(\theta,\phi)\right|^2}_{\text{element pattern}} \times \underbrace{F_a(\theta)}_{\text{array factor}}$$

Note: we are implicitly assuming that the maximum values of the element pattern and the array factor are both one.

- The array factor is the far-field (normalized) radiation intensity of the N array elements were they isotropic radiators.

Array Factor

$$F_a(\theta) = \left| \sum_{i=0}^{N-1} a_i e^{j\psi_i} e^{jikd\cos\theta} \right|^2$$

- The amplitude coefficients (a_i) determine the shape of the pattern (beamwidth, sidelobes, etc.) while the phase coefficients (ψ_i) determine the direction of maximum radiation.

Uniform Array

- A ***uniform array*** is an array of identical elements each fed with an excitation of identical amplitude and progressive phase.
- Such arrays can be used to provide steering of the main beam using electronic scanning realizing an ESA (Electronically Scanned Array).

Array Factor for Uniform Array (1 of 3)

Progressive element-to-element phase shift

$$a_i = 1$$
$$\psi_i = -i\delta$$

$$\delta = k_0 d \cos\theta_0$$

Scan angle

$$F_a(\theta) = \left| \sum_{i=0}^{N-1} e^{ji(kd\cos\theta - \delta)} \right|^2 = \left| \sum_{i=0}^{N-1} e^{jikd(\cos\theta - \cos\theta_0)} \right|^2$$

$$= \left| \sum_{i=0}^{N-1} e^{ji\gamma'} \right|^2 \qquad \gamma' = k_0 d(\cos\theta - \cos\theta_0)$$

Array Factor for Uniform Array (2 of 3)

- Recall:

$$\sum_{n=0}^{N-1} r^n = \frac{1-r^N}{1-r}$$

- Thus,

$$F_a = \left|\sum_{i=0}^{N-1} e^{ji\gamma'}\right|^2 = \left|\frac{1-e^{jN\gamma'}}{1-e^{j\gamma'}}\right|^2 = \frac{\sin^2\left(\frac{N\gamma'}{2}\right)}{\sin^2\left(\frac{\gamma'}{2}\right)}$$

Array Factor for Uniform Array (3 of 3)

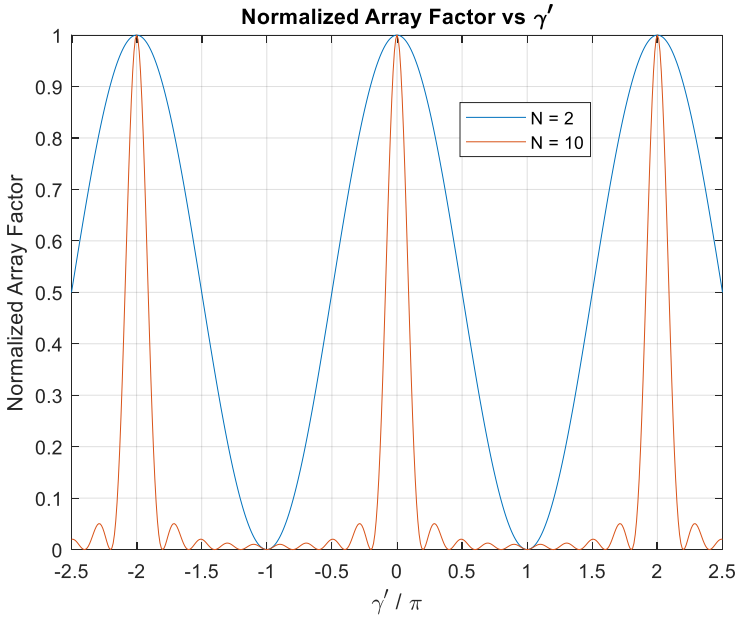
- Evaluating the array factor at γ' = 0 yields its maximum value

$$F_a|_{\gamma'=0} = \lim_{\gamma'\to 0} \frac{\sin^2\left(\frac{N\gamma'}{2}\right)}{\sin^2\left(\frac{\gamma'}{2}\right)} = N^2$$

- Define a normalized array factor as

$$F_{an}(\theta) = \frac{\sin^2\left(\frac{N\gamma'}{2}\right)}{N^2\sin^2\left(\frac{\gamma'}{2}\right)}$$

Normalized Array Factor for N = 2 and N = 10



Peaks of the Array Factor

- Peaks of the array factor occur when both the numerator and denominator of the array factor are equal to zero.
- When the denominator of the array factor is equal to zero, the numerator must also be equal to zero. (But the converse is not true: the numerator can be zero but the denominator is non-zero.)
- The peaks of the array factor correspond to the ***main lobe*** and to ***grating lobes***.

Nulls of the Array Factor

- Nulls of the array factor occur when the numerator of the array factor is equal to zero, but the denominator is not.
- The number of nulls determines the number of sidelobes.

Visible Space and Mapping Between θ and γ'

- What features of the array factor are observable in visible space is determined by the mapping between θ and γ' which depends on the element spacing (kd) and the scan angle (θ_0).
- The range of values for γ' manifesting in visible space correspond to values of θ between 0 and 180 degrees.
- The minimum value of γ' manifesting in visible space corresponds to $\theta = 180^\circ$
- The maximum value of γ' manifesting in visible space corresponds to $\theta = 0$.

Example of θ-to-γ' Mapping (1 of 2)

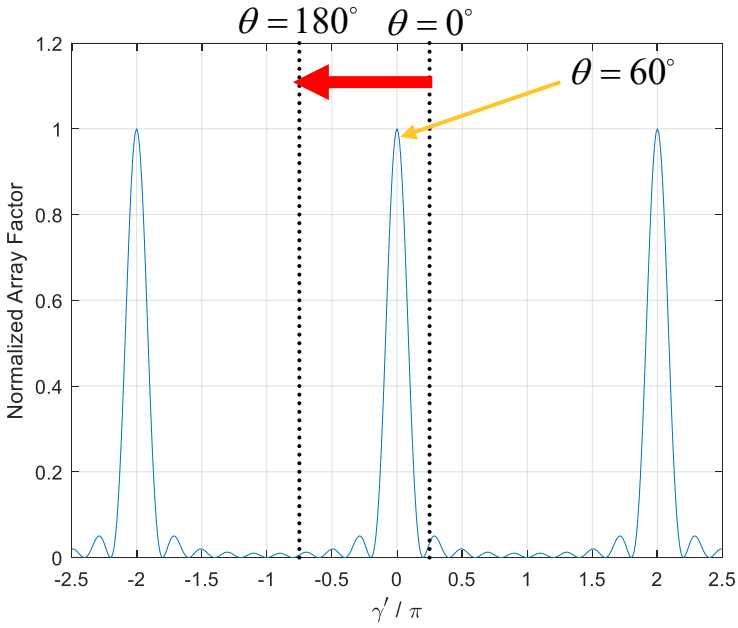
- Let $d = \lambda/4$, $\delta = -\pi/4$, and $N = 10$

$$\theta_0 = \cos^{-1}\left(\frac{-\delta}{kd}\right) = \cos^{-1}\left(\frac{\frac{\pi}{4}}{\frac{\pi}{2}}\right) = \cos^{-1}\left(\frac{1}{2}\right) = 60°$$

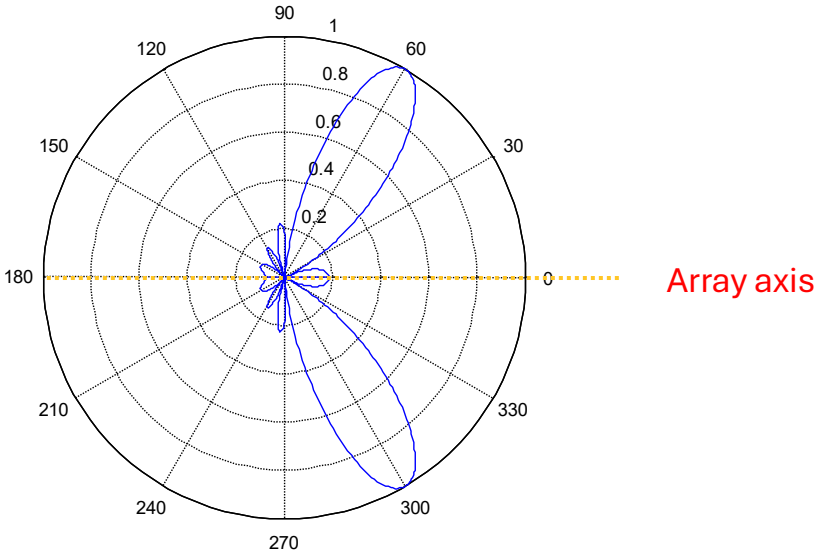
$$\gamma'_{\max} = kd + \delta = \frac{\pi}{4}$$

$$\gamma'_{\min} = -kd + \delta = \frac{-3\pi}{4}$$

Example of θ-to-γ' Mapping (2 of 2)



Example of θ-to-ψ Mapping



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